Computer-Aided Civil and Infrastructure Engineering 48 (2026) 100073

Contents lists available at ScienceDirect

Computer-Aided Civil and Infrastructure Engineering

journal homepage: www.sciencedirect.com/journal/computer-aided-civil-and-infrastructure-engineering

Research Article

Topology guided fourier neural operator with decoupled geometric parameterization for dam leakage detection

Chaoheng Yin [a,b], Yue Liang [a,*], Xiao Pan [b,c,d,**], Bin Xu [a], Bin Zhang [a]

[a] National Engineering Research Center for Inland Waterway Regulation, Chongqing Jiaotong University, Chongqing 400074, China
[b] Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong SAR, China
[c] Guangzhou HKUST Fok Ying Tung Research Institute, Guangzhou, China
[d] HKUST Shenzhen - Hong Kong Collaborative Innovation Research Institute, Shenzhen, China

ARTICLE INFO

Keywords:
Electrical resistivity tomography
Operator learning
Leakage detection
Earth dams
Safety monitoring of infrastructure

ABSTRACT

Traditional electrical resistivity tomography (ERT) inversion for dam leakage detection faces heavy computational loads and smoothness priors that fail to localize anomalies. This study introduces the Topology-Guided Fourier Neural Operator framework. A Fourier Neural Operator replaces the numerical solver based on the governing equations, speeding up forward potential field simulations. To easily locate anomalies, a decoupled geometric parameterization separates the deterministic defect from the stochastic background. A Navigator network extracts coarse topological priors to initialize a parallel multi-hypothesis workflow. Tested across multiple trials, the framework achieves a 99.7% average success rate and an 89.3% high-fidelity reconstruction rate. It accelerates forward modeling by 345 times and overall inversion by 72 times compared to finite element methods. The method detects unseen low-contrast early leakages and remains stable under 20% measurement noise and 50% data loss. This approach provides the efficiency and reliability requisite for routine safety monitoring and operational digital twins.

1. Introduction

Internal leakage represents a leading factor in earth dam failures, a risk exacerbated by increasingly extreme hydro-climatic events (ICOLD, 2019; Moreno-Rodenas et al., 2025; Nasser et al., 2025). While methods such as Ground Penetrating Radar (GPR) (Bekele et al., 2024; Dong et al., 2020) and Distributed Optical Fiber Temperature Sensing (DTS) (Bekele et al., 2024, 2025) characterize internal structures effectively, they are operationally complex for routine monitoring. Electrical Resistivity Tomography (ERT) offers a practical alternative. Electrical conductivity is a sensitive indicator of structural integrity (Guo et al., 2022; Xiao et al., 2024; Zhang et al., 2026), highly responsive to moisture variations and leakage. The technique operates by injecting direct current through source electrodes and measuring the resulting potential differences to reconstruct the internal conductivity distribution. Reconstructing this distribution requires solving a highly non-linear, ill-posed inverse problem constrained by the Poisson equation (Loke & Barker, 1996). Conventional methods typically enforce smoothness constraints to stabilize the inversion (Constable et al., 1987; Liu et al., 2021). However, this regularization tends to smooth out the sharp boundaries of early-stage leakage and masks high-contrast anomalies. Despite recognition of this topological mismatch (Chan & Tai, 2004; Ishizu et al., 2025), accurately resolving such distinct targets within heterogeneous backgrounds remains difficult, especially when using reduced-order parameterizations.

To mitigate the computational burden of stochastic inversion for earth dams, recent research has focused on two main directions: algorithmic acceleration and surrogate modeling. Fast direct solvers expedite Jacobian matrix calculations (Ambikasaran et al., 2013; Saibaba & Kitanidis, 2012); however, high-fidelity probabilistic characterization remains expensive due to the cumulative demand of repeated forward simulations. Standard surrogate modeling frameworks project high-dimensional parameters onto truncated linear subspaces, often employing the Karhunen-Loève (KL) expansion (Ghanem & Spanos, 1991) combined with shallow regressors such as Gaussian Processes (Rasmussen & Williams, 2006). Nevertheless, dependence on global

* Corresponding author at: National Engineering Research Center for Inland Waterway Regulation, Chongqing Jiaotong University, Chongqing 400074, China.
** Corresponding author at: Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, SAR Hong Kong, China.
E-mail addresses: liangyue2560@163.com (Y. Liang), cexiaopan@ust.hk (X. Pan).

https://doi.org/10.1016/j.cacaie.2026.100073
Received 12 January 2026; Accepted 29 March 2026
Available online 11 May 2026
1093-9687/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

basis functions presents limitations. The Kolmogorov n-width exhibits a slow decay rate in these scenarios (Lee & Carlberg, 2020). Such linear projections fail to resolve sharp and high-contrast discontinuities found in internal leakage, inevitably blurring localized anomalies and compromising topological fidelity in reduced-dimensional spaces.

Deep learning provides a powerful tool for inverse modeling and structural health assessment. Recent studies demonstrate that data-driven models successfully evaluate structural conditions and predict physical responses, even under data scarcity or unsupervised settings (Amezquita-Sanchez et al., 2018; Entezami et al., 2026; Javadinasab Hormozabad et al., 2021; Rafiei & Adeli, 2018). While standard network architectures are effective for parameter estimation (Quqa et al., 2023; Taghizadeh et al., 2025; Xue et al., 2025; Zhang et al., 2026), they map between fixed grids, making their performance heavily dependent on the grid density used during training. This dependency limits generalization capabilities (Cha et al., 2024; Zhang et al., 2023). Raissi et al. (2019) proposed Physics-Informed Neural Networks (PINNs) to incorporate physical constraints by embedding partial differential equation (PDE) residuals into the optimization objective. PINNs perform effectively in various fields (Guo et al., 2023; Haghighat et al., 2021; Ji et al., 2024). Yet, because they function as solvers for specific instances rather than learning the general solution operator, each new inversion task mandates retraining from scratch. Furthermore, optimization frequently fails from ill-conditioned loss landscapes (Karniadakis et al., 2021; Krishnapriyan et al., 2021; Wang et al., 2021). Operator learning frameworks extend neural networks to infinite-dimensional function spaces. Prominent examples include the Fourier Neural Operator (FNO) (Li et al., 2021) and spectral graph networks (Yin et al., 2026). The FNO efficiently captures the global dependencies of the steady-state Poisson equation by parameterizing the integral kernel within the frequency domain. This spectral convolution ensures resolution-invariant forward modeling, overcoming the grid-dependence of standard convolutional networks (Kaewnuratchadasorn et al., 2024; Kovachki et al., 2023). Applying these operators to ill-posed inverse problems remains challenging. Most existing neural-operator frameworks execute a direct, unconstrained inverse mapping from measurements to physical properties (Long et al., 2025). This unparameterized approach inherently diffuses sharp topological boundaries under the severe ill-posedness of ERT (Molinaro et al., 2023). While classical PDE optimization extensively employs geometric parameterizations to maintain sharp interfaces, integrating such explicit structural priors into data-driven neural operator frameworks remains underexplored, particularly for the severely ill-posed ERT inverse problem. The absence of topological priors leaves gradient-based optimization vulnerable to local minima, causing failures in identifying localized leakage anomalies.

Within this context, a novel Topology-Guided Fourier Neural Operator (TG-FNO) framework is presented for leakage anomaly detection in earth dams. This framework integrates a decoupled geometric parameterization to disentangle anomalies from stochastic soil backgrounds. A lightweight Guided Network extracts coarse geometric priors directly from electrical resistivity tomography (ERT) measurements to initialize the inversion. Driven by a differentiable FNO surrogate, this topology-guided workflow enables efficient gradient-based optimization, bypassing the computational bottlenecks of conventional finite element solvers. The overall architecture of the TG-FNO framework is shown in Fig. 1.

The primary methodological contributions are:

1. Operator-Driven Inversion Framework: The TG-FNO framework extends the application of Fourier neural operators from standard forward modeling to complex geometric inversion, enabling the resolution of anomalies in ERT.
2. Decoupled Geometric Parameterization: A strategy is developed to mathematically separate deterministic leakage defects from the stochastic background. This resolves the inherent conflict between generic smoothness priors and the sharp topology of leakage anomalies.
3. Topology-Guided Optimization: A hierarchical initialization workflow constrains the inversion search space using geometric priors derived from the Navigator. This mitigates the local minima stagnation typical of gradient-based solvers without requiring expensive global search algorithms.
4. Fast Differentiable Surrogate: An end-to-end differentiable pipeline is established by replacing iterative finite element modeling with the

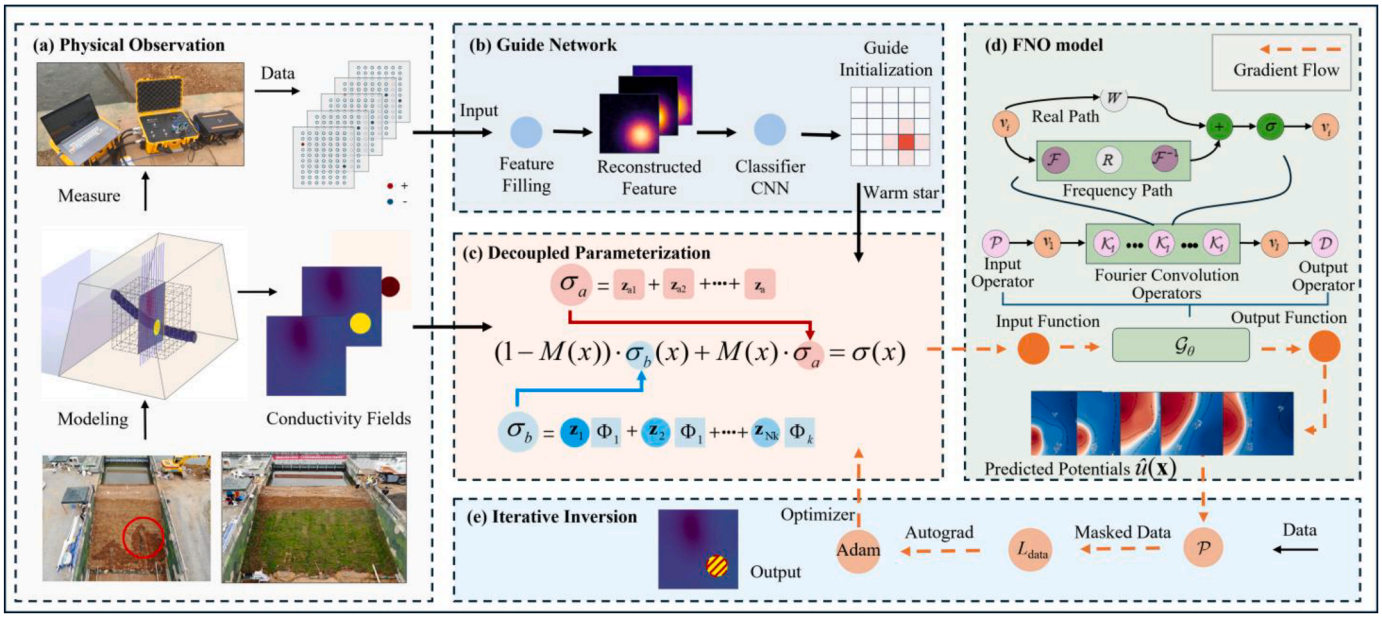


Fig. 1. Architecture of the TG-FNO framework for dam leakage anomaly detection. (a) Physical observation module illustrating the abstraction from real-world dam scenarios to discrete ERT potential measurements; (b) Guided Network designed to infer coarse geometric priors for warm-start initialization; (c) Decoupled geometric parameterization strategy for mathematically disentangling deterministic anomalies from stochastic background heterogeneity; (d) FNO surrogate model serving as a differentiable forward operator; (e) Iterative inversion workflow where the data misfit L_{data} drives the latent variable update via the Adam optimizer and automatic differentiation.

FNO surrogate. This acceleration provides the computational efficiency requisite for routine safety monitoring and operational digital twins.

2. Methodology

The methodology establishes the Topology-Guided Fourier Neural Operator (TG-FNO) framework for electrical resistivity tomography (ERT) inversion. Formulated upon the steady-state ERT governing equations (Section 2.1), the inverse problem is resolved through a differentiable pipeline. A Fourier Neural Operator (FNO) serves as the forward surrogate to accelerate potential field simulations (Section 2.2). A decoupled geometric parameterization is established to disentangle anomalies from the stochastic background (Section 2.3). These components are integrated into a topology-guided inversion workflow, where a Navigator network initializes the gradient-based optimization using coarse geometric priors (Section 2.4).

2.1. Governing equations of steady-state ERT

The ERT forward problem is governed by the steady-state Poisson equation within a bounded domain $\Omega \subset \mathbb{R}^2$. Given a conductivity distribution $\sigma(\mathbf{x})$ and a current source density $I(\mathbf{x})$, the electric potential $u(\mathbf{x})$ satisfies:

$$-\nabla\cdot(\sigma(\mathbf{x})\nabla u(\mathbf{x})) = I[\delta(\mathbf{x}-\mathbf{x}_A) - \delta(\mathbf{x}-\mathbf{x}_B)], \quad \mathbf{x}\in\Omega \tag{1}$$

where $\sigma(\mathbf{x})$ denotes electrical conductivity (S/m); $u(\mathbf{x})$ denotes the electric potential (V); and I (A) represents current intensity. Source and sink terms are modeled via Dirac delta functions $\delta(\cdot)$ at injection location $\mathbf{x}_A$ and extraction location $\mathbf{x}_B$. The domain boundaries $\partial\Omega$ are divided into Dirichlet (Γ_D) and Neumann (Γ_N), satisfying:

$$\begin{cases} u(\mathbf{x}) = 0, & \mathbf{x}\in\Gamma_D \\ \sigma(\mathbf{x})\dfrac{\partial u}{\partial \mathbf{n}} = 0, & \mathbf{x}\in\Gamma_N \end{cases} \tag{2}$$

where $\mathbf{n}$ denotes the unit vector normal to the surface, let A be the non-linear forward operator mapping conductivity to the potential field. The discrete observation vector $\mathbf{d}_{obs}\in\mathbb{R}^m$ contains m potential measurements. Accounting for uncertainty, the observation equation is defined as:

$$\mathbf{d}_{obs} = \mathscr{A}(\mathscr{F}(\sigma)) + \epsilon \tag{3}$$

where $\epsilon\in\mathbb{R}^M$ represents additive noise in the measurements, this noise follows a zero-mean Gaussian process.

2.2. Fourier neural operator as forward surrogate

To accelerate potential field calculations, a data-driven surrogate $\mathscr{G}_\theta$ based on the FNO architecture replaces the conventional numerical solver. This surrogate approximates the mapping from log-conductivity $a(\mathbf{x}) = \ln\sigma(\mathbf{x})$ to the potential field $u(\mathbf{x})$. The architecture comprises a lifting layer, stacked spectral integration layers, and a decoding layer. The input $a(\mathbf{x})$ is concatenated with the spatial coordinate grid. A lifting layer P, implemented as a pointwise linear transformation, projects this input to a high-dimensional latent representation $v_0(\mathbf{x})$:

$$v_0(\mathbf{x}) = P(a(\mathbf{x}), \mathbf{x}) \tag{4}$$

The latent features v_0 propagate through L Fourier layers. The update rule at layer l sums a local linear transform W_l and a global spectral kernel $\mathscr{K}_l$:

$$v_{l+1}(\mathbf{x}) = \rho(W_l v_l(\mathbf{x}) + (\mathscr{K}_l v_l)(\mathbf{x})), \quad l = 0,\ldots,L-1 \tag{5}$$

where $\rho(\cdot)$ denotes the non-linear activation, the system computes the kernel operator $\mathscr{K}_l$ within the frequency domain via the FFT algorithm:

$$(\mathscr{K}v_l)(\mathbf{x}) = \mathscr{F}^{-1}(R_l\cdot\mathscr{F}(v_l))(\mathbf{x}) \tag{6}$$

where $\mathscr{F}$ and $\mathscr{F}^{-1}$ represent the forward and inverse Fourier transforms. The term R_l stands for a trainable complex-valued tensor. Finally, a decoding network Q projects the features v_L to the multi-channel output potential field:

$$\mathbf{u}(\mathbf{x}) = Q(v_L(\mathbf{x})), \quad \mathbf{u}(\mathbf{x})\in\mathbb{R}^{N_s} \tag{7}$$

where N_s is the number of source configurations. This design enables simultaneous reconstruction of potential fields for all sources in a single forward pass, preserving discretization invariance. Parameter updates for θ are driven by the reduction of the mean relative L_2 error across the training dataset. The objective function is:

$$\mathscr{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\frac{\|\mathbf{u}_i - \hat{\mathbf{u}}_i\|_2}{\|\mathbf{u}_i\|_2} \tag{8}$$

where $\mathbf{u}_i$ is the ground truth potential field from the FEM solver, $\hat{\mathbf{u}}_i = \mathscr{G}_\theta(a_i)$ is the predicted field, and $\|\cdot\|_2$ denotes the L_2 norm over the spatial domain and channels.

2.3. Decoupled geometric parameterization

To represent high-contrast localized anomalies within a heterogeneous background, the parameter space is divided into a stochastic background component and a deterministic anomaly component. A unified latent vector is defined as $\mathbf{z} = [\mathbf{z}_{bg}, \mathbf{z}_a]^\top$, where $\mathbf{z}_{bg}$ and $\mathbf{z}_a$ control the background heterogeneity and the anomaly characteristics, respectively. The conductivity field $\sigma(\mathbf{x})$ is constructed via a differentiable composite model:

$$\sigma(\mathbf{x};\mathbf{z}) = [1 - M(\mathbf{x};\mathbf{z}_a)]\cdot\exp\big(Y(\mathbf{x};\mathbf{z}_{bg})\big) + M(\mathbf{x};\mathbf{z}_a)\cdot\sigma_{val} \tag{9}$$

where $Y(\mathbf{x})$ is the background log-conductivity field, and σ_{val} is the conductivity magnitude of the anomaly. The term $M(\mathbf{x})\in[0,1]$ serves as a topology mask. It defines the spatial extent of the anomaly. This formulation yields sharp interfaces. It also preserves the differentiability required for gradient-based inversion. A Gaussian Random Field (GRF) describes the background heterogeneity to capture spatial correlations. The truncated Karhunen-Loève (KL) expansion parameterizes $Y(\mathbf{x})$ (Tang et al., 2020; Zhang & Lu, 2004):

$$Y(\mathbf{x};\mathbf{z}_{bg}) = \mu(\mathbf{x}) + \sum_{i=1}^{N_k}\sqrt{\lambda_i}\phi_i(\mathbf{x})z_i \tag{10}$$

In this equation, $\mu(\mathbf{x})$ denotes the mean field. The terms λ_i and ϕ_i denote the principal eigenvalues and eigenfunctions derived from the covariance kernel, respectively. The vector $\mathbf{z}_{bg} = [z_1, \ldots, z_{Nk}]^\top$ represents the stochastic coefficients following a standard normal distribution. The geometry of the anomaly is stored in the latent vector $\mathbf{z}_a$. This geometry is defined via an implicit potential function $\Psi(\mathbf{x};\mathbf{z}_a)$, where the interface corresponds to the zero-level set $\{\mathbf{x}\in\Omega \mid \Psi(\mathbf{x}) = 0\}$. A radial basis formulation is adopted to parameterize the potential:

$$\Psi(\mathbf{x};\mathbf{z}_a) = r^2 - \|\mathbf{x}-\mathbf{c}\|_2^2 \tag{11}$$

where $\mathbf{z}_a = [c^\top, r^\top]^\top$ contains the center and radius parameters, while initial dam leakage is irregular, internal erosion typically expands radially. Geotechnical practice models concentrated leak erosion using an effective radius (Bonelli & Brivois, 2008; Wan & Fell, 2004). Additionally, volumetric diffusion in macroscopic field measurements blurs complex defect boundaries into roughly circular signatures (Bherde et al., 2025; Xiao et al., 2024). Parameterizing the anomaly as an equivalent circle prevents over-parameterization of the inverse problem. This geometry directly captures the location and scale necessary for early safety warnings. To allow for gradient-based optimization, the

sharp interface is relaxed into a continuous transition via a logistic sigmoid mapping:

$$M(\mathbf{x};\mathbf{z}_a)=\frac{1}{1+\exp(-\eta\cdot\Psi(\mathbf{x};\mathbf{z}_a))} \tag{12}$$

where $\eta>0$ is a scalar hyperparameter controlling interface sharpness. As $\eta\to\infty$, the mask approaches a Heaviside step function. This differentiable formulation allows gradients to backpropagate from the loss function to the geometric latent variables.

2.4. Topology-guided inversion workflow

The inversion process performs a deterministic search for the latent code $\mathbf{z}$ to minimize the discrepancy between measured and predicted potentials. Geometric parameterization is highly sensitive to initial conditions. Therefore, a parallel multi-hypothesis strategy is used to ensure robust convergence. A candidate set $\mathcal{Z}=\{\mathbf{z}_k\}$ $(k=1\ldots N_h)$ is defined, where N_h denotes the total number of parallel chains. The initialization of each trajectory uses priors derived from the Navigator network, $\mathcal{N}_\phi$, combined with stochastic perturbations:

$$\mathbf{z}_k^{(0)}=\mathcal{T}(\mathbf{h}_k)+\boldsymbol{\xi}_k \tag{13}$$

Here, $\mathbf{h}_k$ is the k-th topological prior. It is extracted as the discrete spatial index corresponding to the high-probability regions in the Navigator's output map. The operator $\mathcal{T}$ maps these discrete grid locations into the continuous latent manifold. The term ξ_k represents random perturbations sampled from a defined probability distribution to encourage local exploration around the identified modes. Upon initialization, a gradient-based algorithm optimizes all candidates in $\mathcal{Z}$ simultaneously. The optimization process targets the minimization of the L_2 data misfit for each hypothesis independently:

$$\mathcal{J}(\mathbf{z})=\|\,\mathbf{d}_{\text{obs}}-\Pi(\mathcal{G}_\theta(\ln\sigma(\cdot;\mathbf{z})))\,\|_2^2 \tag{14}$$

where Π is the projection and masking operator mapping the predicted continuous potential field to the discrete electrode observations, implemented as a fixed binary tensor, Π assigns a value of zero to the active injection and extraction electrodes. All remaining passive measurement entries retain a value of one. Gradients are computed via backpropagation through $\mathcal{G}_\theta$, allowing the parallel chains to converge to their respective local optima. The final solution $\hat{\mathbf{z}}$ corresponds to the candidate yielding the minimum residual error:

$$\hat{\mathbf{z}}=\underset{\mathbf{z}\in\mathcal{Z}}{\operatorname{argmin}}\mathcal{J}(\mathbf{z}) \tag{15}$$

3. Numerical experiments

3.1. Dataset generation and experimental setup

The numerical setup involved a 2D vertical dam profile Ω measuring 20 m × 20 m. A mesh of 128 × 128 uniform quadrilateral cells covers this domain. Data acquisition simulated a cross-borehole configuration with a 7 × 10 electrode array. The array yields horizontal and vertical sensor spacings of 3.0 m and 2.0 m, respectively. Anomalies smaller than this spacing generate weak electrical signatures. These limited potential variations are readily masked by background heterogeneity, inherently restricting the minimum detectable size. The conductivity field $\sigma(\mathbf{x})$ combines a stochastic background with a discrete anomaly. This approach allows for independent control over background heterogeneity and anomalous features, aligning with the decoupled inversion framework. The Karhunen-Loève (KL) expansion represented the background heterogeneity as a Log-Normal Gaussian Random Field (GRF). Key statistical settings defined the mean conductivity at 0.005S/m and the standard deviation at 0.002S/m. The correlation length was 5.0 m. To simulate seepage, a high-conductivity circular anomaly was introduced, parameterized by its centroid $\mathbf{c}=(c_x, c_y)$ and radius r. The centroid was confined to the central region [5.0, 15.0] m to reduce boundary effects, with the radius varying between 1.0 m and 3.0 m. A log-normal distribution determined the anomaly conductivity. The mean value stood at approximately 0.3 S/m. A sigmoid transition mask smoothed the interface to maintain numerical stability. Latin Hypercube Sampling (Helton & Davis, 2003) generated 1000 independent samples, split into 900 for training and 100 for testing. Forward modeling was performed using the SfePy finite element library. To ensure a unique solution for the Neumann problem, a single-node Dirichlet reference ($u=0$) was enforced at the vertex (0,0). This anchor serves strictly as a mathematical gauge-fixing condition. The Neumann boundary value problem for the forward model is singular without a defined reference potential. The anchor ensures the convergence of the finite element solver. Relocating this anchor to an alternative node introduces a global constant shift to the computed potential field. However, the observational inputs for the inversion framework consist exclusively of potential differences between electrode pairs. The constant shift perfectly cancels out during the calculation of these differences. Therefore, the spatial placement of the Dirichlet anchor exerts no influence on the final inversion performance. A "5-stim" protocol was employed (see Fig. 2). The acquired potentials were log-transformed to compress the dynamic range. Key parameters are listed in Table 1. All numerical simulations were run on a standard desktop workstation powered by an Intel Core i7–12,700 K CPU 4.9 GHz. The framework was implemented in Python 3.12.3, using the SfePy library (version 2025.1) for finite element analysis (Cimrman et al., 2019).

3.2. Computational implementation

3.2.1. Training of FNO surrogate and topology navigator

The computational environment for all neural network training relied on the PyTorch framework, with hardware acceleration provided by a workstation equipped with an NVIDIA GeForce RTX 4070 Super GPU. The FNO surrogate maps the log-conductivity field $\ln\sigma(x)$ to the log-transformed potential fields. Exploiting the resolution-invariance of

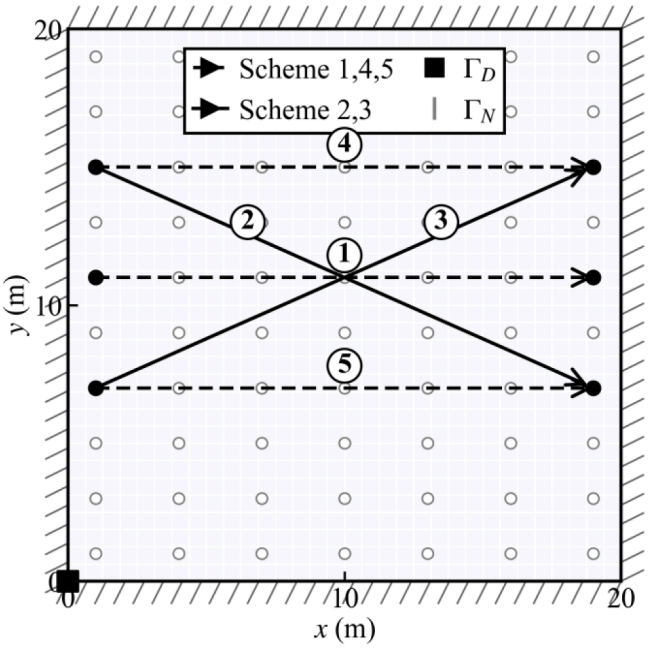


Fig. 2. Illustration of the simulation geometry and experimental configuration. The 20 m × 20 m domain is discretized into quadrilateral finite elements. 7 × 10 electrode array employing five distinct dipole injection schemes (Schemes 1–5). Γ_D denotes the single Dirichlet anchor at the origin ensuring uniqueness; Γ_N denotes insulated external boundaries.

Table 1
Geometric specifications and statistical parameters for the synthetic dataset generation.

Category	Parameter	Value / Range
Domain	Domain Size, Ω	20 m × 20 m
	Grid Resolution, $N_x \times N_Y$	128 × 128
Background	KL Latent Dimension, d_{bg}	20
	Mean Conductivity, μ_{bg}	0.005 S/m
	Std. Deviation, σ_{bg}	0.002 S/m
	Correlation Length, l_c	5.0m
Anomaly	Latent Dimension, d_a	4
	Position Range, c	[5.0, 15.0] m
	Radius Range, r	[1.0, 3.0] m
	Conductivity, σ_a	≈ 0.3 S/m(Mean)

operator learning, training was performed on a subsampled 64 × 64 grid. This reduces computational cost while allowing the learned operator to be evaluated on the native 128 × 128 mesh or arbitrary resolutions without retraining. The FNO architecture consists of four Fourier spectral layers ($L = 4$) with a channel width of 32, preserving the lowest 12 frequency modes ($k_{\max}$=12). A padding ratio of 0.1 was applied to reduce boundary artifacts. Parameter optimization was performed using the Adam algorithm, initialized with a learning rate of 1×10^{-3} that decayed according to a cosine annealing schedule. Gaussian noise with a standard deviation of 0.05 was injected into the ground-truth potential data. The FNO training process was completed in approximately 508 s. The GuideNet locates coarse anomalies using sparse electrode measurements. The input is a 5 × 10 × 7 tensor representing potential data across 5 stimulation schemes on the 7 × 10 electrode array. The network includes a Feature Reconstruction block that upsamples these sparse inputs, followed by a classification block to localize the anomaly within a specific grid resolution. Odd-numbered grids of 3 × 3, 5 × 5, and 7 × 7 are evaluated to maintain a geometric center cell that aligns with typical internal defect distributions. The 3 × 3 grid achieves 80.0% Top-1 accuracy but yields sub-regions too coarse for effective downstream gradient optimization. The 7 × 7 grid creates an overly fragmented classification space for the initial guess. Adjacent grid cells produce highly similar boundary signatures. This similarity increases class confusion, reduces the Top-1 accuracy to 65.0%, and exacerbates mislocalization errors. The 5 × 5 grid yields optimal performance, achieving 81.0% Top-1 and 99.0% Top-3 accuracies. This resolution balances the spatial sensitivity limits of the measurements with the requirement for a constrained bounding area. The dual-scale initialization strategy applied during the subsequent inversion phase explicitly mitigates potential mislocalization cases at this chosen resolution. For this task, the AdamW solver was employed to update the weights, configured with a constant learning rate of 1×10^{-3}. Random Gaussian noise of the same magnitude was added to the input potentials to simulate sensor noise. Training finished in approximately 23 s.

3.2.2. TG-FNO inversion configuration

The inverse problem is formulated as an optimization task, wherein the objective is to minimize the discrepancy between the acquired electrode measurements and the potential fields predicted by the surrogate. Gaussian noise is added directly to the log-transformed potentials to simulate measurement uncertainty. This additive operation in the logarithmic domain equates to applying multiplicative log-normal noise in the linear domain. This approach physically models instrumentation errors that scale with signal magnitude. Measurements coinciding with active source and sink electrodes are excluded to avoid singularity artifacts. To ensure robustness against local minima, a topology-guided parallel search strategy is adopted. Specifically, the number of parallel chains N_h is set to 16. These chains execute simultaneously via GPU batch processing. A dual-scale initialization strategy is implemented to mitigate potential misguidance from the neural prior under severe noise. Among the 16 parallel chains, five are instantiated at the Top-5 probability predictions extracted from the GuideNet as the topological priors $\mathbf{h}_k$. These five chains are coupled with a standard exploration jitter ξ_k sampled from $(0, 0.05^2)$ within the normalized latent space. To compensate for guide mismatch, the remaining 11 chains are initialized around the Top-1 prediction but are subjected to a fourfold larger exploration jitter sampled from $(0, 0.20^2)$. This configuration balances the exploitation of topological priors with latent space, preventing the optimization process from being trapped in misleading local minima. The dimension of the background latent vector d_{bg} is set to 20. This truncation threshold is determined through an eigenvalue energy spectrum analysis of the covariance kernel. For the baseline correlation length of 5.0 m, the first 20 modes capture 99.20% of the total variance. Incorporating additional dimensions increases the optimization burden without meaningfully altering the physical representation of the background field. The background latent variables $\mathbf{z}_{\text{bg}}$ are randomly initialized from a normal distribution $(0, 0.1^2)$. The latent vectors are updated deterministically over a fixed 100 iterations via the Adam optimizer. To synchronize the convergence of background and anomaly features, distinct learning rates of 0.01 and 0.05 are assigned to $\mathbf{z}_{\text{bg}}$ and $\mathbf{z}_{\text{a}}$, respectively. The optimization process omits early stopping mechanisms, explicit gradient clipping, and supplementary regularization terms. The geometric parameterization architecture inherently enforces the physical bounds of the conductivity field. The final reconstruction is selected from the optimization trajectory yielding the lowest data residual. This data-driven inversion is highly efficient, achieving an average runtime of approximately 2.39 s per case on the utilized computational platform.

3.3. Quantitative metrics

The predictive accuracy of the forward surrogate is evaluated using the relative L_2 residual error, following the formulation in Eq. (8). The quality of the inverted conductivity fields is assessed via the Structural Similarity Index (SSIM) (Wang et al., 2004), which measures the structural fidelity of the reconstruction against the ground truth. To evaluate the detection capability regarding leakage anomalies, the Intersection over Union (IoU) is used:

$$\text{IoU} = \frac{|\widehat{M} \cap M|}{|\widehat{M} \cup M|} \tag{16}$$

where M denotes the predicted topology mask, based on the IoU, two threshold-based metrics are defined to quantify detection reliability: Acc_0, representing the Inversion Success Rate (percentage of samples satisfying IoU > 0), and $Acc_{0.5}$, representing the High-Fidelity Detection Rate (percentage satisfying IoU > 0.5).

4. Results and discussion

4.1. Predictive performance of FNO surrogate

The FNO surrogate was evaluated on an independent validation set of 100 samples. Fig. 3 shows forward modeling results for five representative realizations (R1–R5). The model accurately maps conductivity fields (Fig. 3A1–A5) to electric potential distributions (Fig. 3C1–C5) spanning diverse anomaly scenarios. It reproduces the equipotential line distortions caused by leakage anomalies relative to the reference (Fig. 3B1–B5 vs. C1-C5). Closer inspection of R4 (comparing Fig. 3B4, and C4) shows that while the anomaly center is correctly localized, the circular geometry of the reference is distorted into an approximate ellipse. This geometric distortion results from spectral truncation in the FNO architecture, where retaining only dominant low-frequency modes acts as a low-pass filter. Minor fluctuations near singularities and domain boundaries are caused by the training strategy, where Gaussian noise was added to target potentials for regularization. The model demonstrated high global accuracy. It recorded a mean relative L_2 error

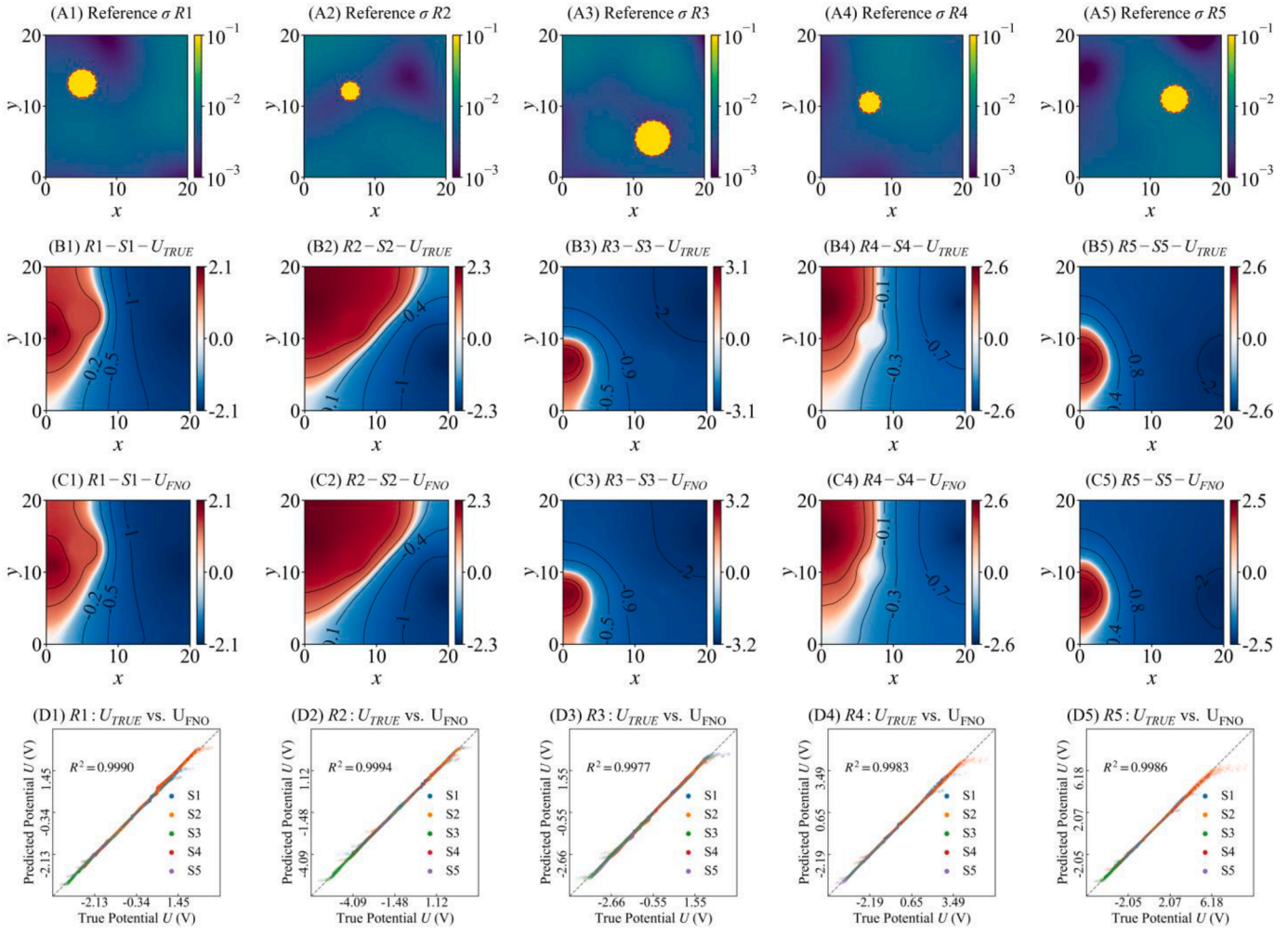


Fig. 3. Performance of the FNO surrogate across various conductivity samples and injection protocols. (A1–A5) Panels display conductivity distributions σ, labeled *R*1 through *R*5; (B1–B5) Reference or true electric potentials U_{TRUE} solved by FEM under corresponding injection schemes (*S*1–*S*5), for example, *R*1–*S*1–U_{TRUE} denotes the potential field of realization *R*1 under injection scheme *S*1; (C1–C5) Electric potentials predicted by FNO U_{FNO} for the same realizations and schemes; (D1–D5) Scatter plots of U_{TRUE} versus U_{FNO} for each realization including data from all five injection schemes, showing R^2coefficients consistently exceeding 0.99.

of 3.18%. Additionally, the coefficient of determination R^2 reached 0.9978 across the validation set. A channel-wise analysis verified robustness across different current injection patterns. Performance was consistent across all five protocols (Schemes 1–5), with relative errors of 3.37% ± 1.41%, 3.19% ± 1.20%, 3.28% ± 1.82%, 3.28% ± 1.05%, 2.80% ± 1.43%, respectively. Scatter plots (Fig. 3D1–D5) support this consistency: the alignment of clusters (S1–S5) along the identity line shows that accuracy is stable regardless of the injection scheme. Finally, the FNO model reduces average inference time from 0.8287 s to 0.0024 s per simulation. This 345 × speedup removes the computational bottleneck for subsequent stochastic inversion, which requires extensive forward evaluations.

4.2. Inversion performance of TG-FNO framework

4.2.1. Baseline selection

The proposed framework is benchmarked against two baselines to evaluate its inversion performance and isolate architectural contributions. The first baseline is a conventional stochastic inversion combining the finite element method (FEM) with the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Alkayem et al., 2018; Marty et al., 2023; Tiprak et al., 2025). Preliminary evaluations detailed in Appendix A justify the necessity of this derivative-free global optimizer. Standard continuous inversion and unconstrained generative models struggle with the extreme sparsity of cross-borehole measurements. As illustrated in Fig. A1 and quantified in Table A1, Total Variation-regularized pixel-wise inversion produces over-smoothed conductivity fields, reducing the high-fidelity localization metric $Acc_{0.5}$ to near zero. Furthermore, applying the Gauss-Newton algorithm directly to the decoupled parameterization leads to severe local minima entrapment, while a Conditional GAN introduces background artifacts. Given the poor reconstruction metrics of these local gradient algorithms and data-driven approaches, the robust global search capacity of CMA-ES provides a suitable conventional benchmark. The second baseline is a Random FNO configuration. This internal baseline shares the identical frozen surrogate and continuous optimization scheme as the proposed framework, differing only by relying entirely on random latent initialization to explicitly isolate the performance contribution of the guide network.

4.2.2. Inversion results

Fig. 4 illustrates the qualitative stability of TG-FNO (A1–A5). By explicitly optimizing the decoupled geometric latent variables, the framework consistently resolves the anomaly locations. The alignment between the reconstructed geometries and the ground truth confirms that the topology-guided initialization prevents the optimization from trapping in local minima. In contrast, the baseline FNO (Fig. 4B1–B5) demonstrates high sensitivity to initialization. This disparity in

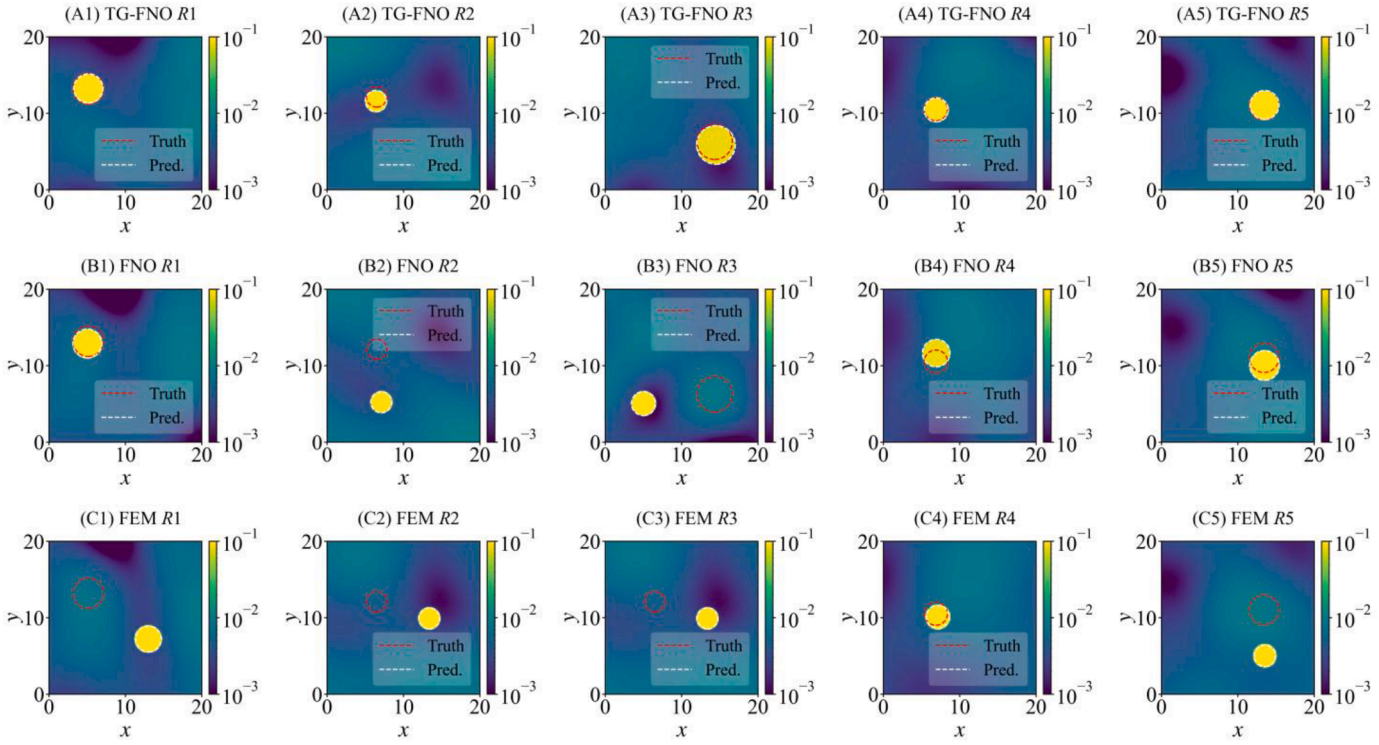


Fig. 4. Comparative inversion results of conductivity fields across five realizations (*R*1–*R*5). (A1–A5) Inverted fields using the proposed TG-FNO framework; (B1–B5) Inverted fields using the FNO surrogate with random initialization; (C1–C5) Inverted fields using conventional stochastic FEM inversion; red dashed contours denote the True Anomaly, while white dashed contours denote the Reconstructed anomaly.

localization success stems from the dependence of gradient-based solvers on the initial guess. In failure cases, the optimization is trapped in local minima, yielding spurious anomalies or incorrect localizations. The FEM approach (Fig. 4C1–C5) frequently fails to converge to the geometric parameters. Despite high computational costs, the amplification of measurement noise results in detection failures (C1, C2, C5) or positional deviations (C3). Quantitative analysis via success rate curves (Fig. 5) confirms this performance gap. The TG-FNO curve maintains a stable high success rate across a wide range of IoU thresholds. Specifically, TG-FNO achieves an Inversion Success Rate Acc_0 of 100%, eliminating the failures observed in the baseline FNO (Acc_0=95%) and the FEM method (Acc_0=76%). Regarding high-fidelity reconstruction, TG-FNO maintains an $Acc_{0.5}$ of 91%, surpassing the baseline FNO ($Acc_{0.5}$ = 82%) and the FEM ($Acc_{0.5}$ = 69%). Beyond localization accuracy, the overall structural fidelity was assessed using the Structural Similarity Index (SSIM). As detailed in Table 2, TG-FNO achieves the highest SSIM of 0.9527, indicating superior preservation of global texture. Notably, while the FEM method achieves a comparable SSIM of 0.9410, its significantly lower Mean IoU of 0.4813 reflects a critical deficiency in precise anomaly segmentation. Repeated trials across multiple random seeds confirm the statistical robustness and the overall accuracy improvement of the proposed approach. TG-FNO achieves a mean Acc_0 of 99.7% ± 0.6%, outperforming the baseline FNO (93.0% ± 3.5%) and FEM (77.3% ± 6.0%). For high-fidelity reconstruction, TG-FNO reaches a mean $Acc_{0.5}$ of 89.3% ± 2.1%. This demonstrates a clear accuracy gain and higher stability compared to the baseline FNO (79.7% ± 6.8%) and FEM (65.7% ± 5.8%). This statistical consistency confirms that topology-guided initialization mitigates optimization uncertainty. Computationally, TG-FNO requires an average inference time of 2.39 s, matching the 2.38 s of the baseline FNO. Conversely, the traditional FEM approach demands approximately 172.32 s per case. This comparison highlights the efficiency advantage of the proposed framework, which achieves an acceleration of approximately 72 × relative to the conventional stochastic method while delivering superior inversion accuracy. This efficiency is further enhanced by a cross-resolution strategy: the surrogate model is trained at a 64 × 64 resolution and deployed at 128 × 128 during inversion. This approach reduces offline training time while maintaining high-fidelity boundary evaluations.

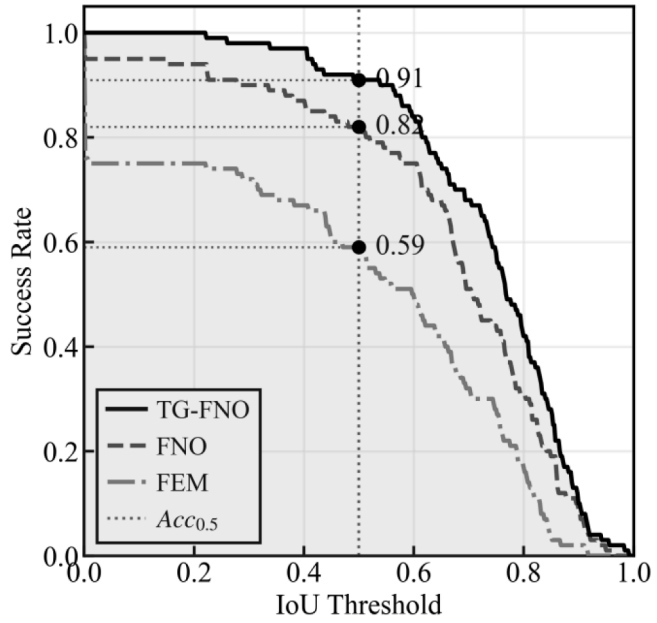


Fig. 5. Quantitative comparison of detection reliability across the full test set. Success rates plotted against IoU thresholds for the proposed TG-FNO, FNO surrogate with random initialization, and conventional stochastic FEM inversion; vertical dotted line marks the high-fidelity detection threshold ($Acc_{0.5}$).

Table 2
Quantitative inversion metrics for the standard baseline and various robustness evaluation scenarios.

Experimental Setting	Evaluation Condition	Mean IoU	Acc_0	$Acc_{0.5}$	SSIM
Standard baseline	TG-FNO	0.7400	100%	91%	0.9527
	FNO random initialization	0.6589	95%	82%	0.9484
	FEM stochastic inversion	0.4813	76%	69%	0.9410
Complex anomaly configurations	Contrast ratio 10	0.6804	100%	82%	0.9474
	Contrast ratio 25	0.7214	100%	89%	0.9528
	Noncircular anomalies	0.6892	100%	100%	0.9418
	Multiple anomalies	0.6866	100%	98%	0.9278
Adverse Monitoring Conditions	Noise level 0.10	0.5903	98%	72%	0.9381
	Noise level 0.20	0.4040	86%	44%	0.9112
	Sensor data loss 25%	0.7401	100%	91%	0.9568
	Sensor data loss 50%	0.6129	99%	75%	0.9430
Prior Mismatch (TG-FNO)	Correlation length 3.0 m	0.6942	97%	87%	0.9406
	Correlation length 7.0 m	0.7407	100%	92%	0.9560

Note: All evaluation conditions incorporate a baseline 0.05 measurement noise level unless explicitly modified in the adverse noise scenarios. Each configuration is evaluated on a set of 100 samples using the pre-trained TG-FNO framework without additional parameter fine-tuning.

4.3. Ablation study

The quantitative metrics for the ablation variants and the complete TG-FNO framework are summarized in Table 3. Replacing the decoupled parameterization with a continuous grid inversion stabilized by total variation regularization reduces the overall accuracy. This pixel-wise baseline maintains an Acc_0 of 78%, but its $Acc_{0.5}$ decreases to 16%, yielding a Mean IoU of 0.2423. This result indicates that directly updating thousands of grid elements under incomplete measurements remains ill-posed. The explicit geometric prior is necessary to stabilize the anomaly segmentation. The initialization and search strategies present a specific optimization trade-off. The random initialization variant executes 16 parallel chains without topological guidance. This stochastic search achieves an $Acc_{0.5}$ of 82%. However, the absence of a spatial prior leads to localization errors in certain cases, resulting in an Acc_0 of 95%. In contrast, the single-chain variant utilizes only the top prediction from the Navigator. This data-driven warm-start guides the optimizer to the general target area and improves the Acc_0 to 98%. Nevertheless, relying on a single trajectory limits local exploration. It traps the solver in suboptimal minima near the true anomaly, which reduces the $Acc_{0.5}$ to 72%. Finally, replacing the FNO surrogate with a conventional physical solver limits the inversion efficiency and accuracy. Evaluated with a derivative-free optimizer, the physical solver yields a Mean IoU of 0.4813. The complete TG-FNO framework integrates all proposed components to achieve a 100% Acc_0 and a 91% $Acc_{0.5}$.

Table 3
Summarizes the quantitative metrics of four model variants evaluated on the validation set to isolate the contribution of each proposed component.

Model Variant	Description	Mean IoU	Acc_0	$Acc_{0.5}$	SSIM
Pixel-wise	Continuous grid inversion with TV regularization	0.2423	78%	16%	0.9232
Random initialization	Optimization without Guide Network warm-start	0.6589	95%	82%	0.9527
Single optimization chain	Top-1 trajectory search without parallel evaluation	0.6154	98%	72%	0.9434
FEM forward solver	Conventional physics-based solver with CMA-ES	0.4813	76%	69%	0.9410
TG-FNO	Complete framework architecture	0.7400	100%	91%	0.9527

4.4. Robustness and generalization analysis

4.4.1. Generalization to complex anomaly configurations

Table 2 evaluates the model generalization on complex anomaly configurations without parameter fine-tuning. The framework is initially tested on low-contrast anomalies representing early-stage leakage with weak electrical signatures. The model trained on a contrast ratio of 60 is directly applied to new test sets with reduced ratios of 25 and 10. The framework maintains a 100% Acc_0 across both groups. For the most challenging 10-fold contrast ratio, it achieves a Mean IoU of 0.6804 and an $Acc_{0.5}$ of 82%. This performance indicates that the FNO surrogate approximates the physical mapping between potential and conductivity instead of memorizing high-contrast training patterns. To evaluate out-of-distribution inversion capabilities, the framework is tested on two independent sets of 100 samples featuring geometric distribution shifts: noncircular elongated profiles and multiple anomalies with varying intensities (Fig. A2). While the training dataset exclusively contains single circular anomalies, the solver adapts to these unseen configurations. In scenarios with multiple defects, the pronounced potential distortion from the primary leakage masks weaker secondary signatures. Constrained by the single-circle parameterization, the model prioritizes the dominant anomaly and reconstructs an equivalent enclosing circle. Consequently, it locates the primary leakage zones with an $Acc_{0.5}$ of 100% for noncircular anomalies and 98% for multiple anomalies. The Mean IoU values of 0.6892 and 0.6866 reflect the geometric difference between the parameterized circle and the actual irregular boundaries. These results confirm the robust out-of-distribution generalization of the topology-guided search against geometric prior mismatches and masking effects.

4.4.2. Robustness to adverse monitoring conditions

To assess noise robustness, the model trained with a 0.05 noise level is directly applied to test data featuring 0.10 and 0.20 noise intensities without parameter fine-tuning. Under the 0.10 noise condition, the framework maintains high accuracy, achieving an Acc_0 of 98% and an $Acc_{0.5}$ of 72%. At the 0.20 noise intensity, the heightened uncertainty reduces the Mean IoU to 0.4040 and the $Acc_{0.5}$ to 44%. However, the framework sustains a substantial detection success rate with an Acc_0 of 86%. This indicates that the explicit geometric parameterization effectively regularizes the inverse problem, facilitating stable anomaly identification even under significant signal interference. Hardware failures and disconnected electrodes are simulated by randomly masking observation entries. With a 25% data loss, the inversion performance remains consistent with the baseline, delivering a Mean IoU of 0.7401 and an Acc_0 of 100%. Under a severe 50% data loss condition, the framework avoids localization failure, achieving a Mean IoU of 0.6129 and an $Acc_{0.5}$ of 75%. An evaluation of this extreme sparsity reveals a size-dependent detection threshold. The reduced sensor density directly constrains the detectable anomaly size. Smaller defects approaching the 2.0-meter theoretical limit fall below the spatial resolution of the degraded array. These targets become difficult to resolve and drive the observed decrease in accuracy. Conversely, larger damages remain accurately located. These results demonstrate the capability of the FNO surrogate and the topology-guided search to reconstruct internal physical fields from highly incomplete measurements. The framework is tested against contact impedance issues, a chronic challenge in field ERT

monitoring that introduces severe non-Gaussian noise. Poor electrode-soil contact often generates extreme outliers that violate standard Gaussian noise assumptions. This specific scenario is simulated by introducing artificial signal amplification (five to ten times the standard magnitude) into three randomly selected electrodes. Standard optimization using an L2 norm (Mean Squared Error) objective fails under these conditions, causing the Acc_0 to drop to 47% due to severe gradient disruption. Because the FNO forward surrogate is entirely decoupled from the inversion objective function, the system easily adapts by switching to an outlier-robust L1 norm (Mean Absolute Error) combined with input signal clipping. This adjustment requires no retraining of the neural networks and successfully neutralizes the abnormal gradients, restoring the Acc_0 to 96% and the Mean IoU to 0.6600. This modularity allows the framework to dynamically accommodate severe local sensor anomalies and non-Gaussian interference during operational monitoring.

4.4.3. Sensitivity to statistical prior mismatch

The final phase of the robustness evaluation examines the sensitivity to statistical prior mismatch. The baseline framework is trained exclusively on random background fields with a spatial correlation length of 5.0 m. This model is directly applied to two independent sets of 100 samples featuring unseen correlation lengths of 7.0 m and 3.0 m without parameter fine-tuning. For the smoother background condition of 7.0 m, the inversion metrics slightly exceed the standard baseline. The model achieves a Mean IoU of 0.7407 and an $Acc_{0.5}$ of 92%. This demonstrates robust out-of-distribution generalization when transferring the model to spatially simpler domains. The 3.0 m correlation length introduces highly fragmented background heterogeneities. These intense spatial variations generate background potential fluctuations that often mask the actual leakage signals. Consequently, the high-fidelity detection rate $Acc_{0.5}$ decreases to 87%, and the Mean IoU drops to 0.6942. The framework nevertheless maintains a 97% Acc_0, successfully isolating the general defect zone despite the severe background noise. These testing outcomes reveal an asymmetric generalization behavior. The surrogate model adapts readily to smoother environments but experiences a slight precision loss in more chaotic physical fields. For practical field deployments, the training dataset should utilize smaller correlation lengths to encompass severe background variations, ensuring optimal downward compatibility across diverse geological conditions. In real-world dam monitoring scenarios, the latent dimension d_{bg} and correlation length l_c strongly dictate the inversion difficulty. These geological priors must not be arbitrarily assumed. They must be established through historical design records and site-specific material sampling. This empirical validation forms a reliable baseline prior to executing the inversion algorithm.

4.5. Computational and practical challenges in field deployment

While transitioning to fully three-dimensional inversion is a logical step for comprehensive dam monitoring, two-dimensional longitudinal profiling remains the main and practical setup for routine, continuous safety assessment of long embankments (Rahimi et al., 2025). Extending the proposed operator learning framework to 3D environments faces significant mathematical and computational challenges, chiefly the curse of dimensionality. The standard three-dimensional FNO architecture increases training time complexity to ($N^3 \log N$), which requires large amounts of memory during network backpropagation on typical engineering hardware (Li & Ye, 2025). Additionally, capturing the spatial variance of a 3D background field requires a vastly expanded latent parameter space. Integrating this expanded representation with 3D geometric coordinates creates a highly complex, non-convex optimization landscape, significantly increasing the probability of local minima entrapment. Maintaining the current sub-second inference efficiency and localization accuracy in 3D necessitates more advanced, memory-efficient operator architectures.

Beyond dimensional extensions, field deployment faces a fundamental simulation-to-reality gap related to boundary conditions. Neural operators approximate Green's functions mathematically, making the surrogate very sensitive to changes in boundary conditions. Achieving out-of-distribution generalization across different cross-sections remains a core challenge. However, operational dam monitoring generally involves static structural profiles and predictable environmental states. Therefore, training a universal operator is unnecessary for this specific use. A practical solution is to pre-train a small set of targeted operators tailored to known seasonal or operational boundary conditions. Using these condition-specific surrogates during field inspections effectively overcomes the boundary mismatch issue at a minimal offline computational cost. The proposed 2D framework provides a computationally efficient and practical foundation for operational digital twins.

5. Conclusion

This study introduces the TG-FNO framework to address the conflict between smoothness regularization and the sharp topological boundaries of leakage anomalies in electrical resistivity tomography. By combining decoupled geometric parameterization with a topology-guided warm start, the method explicitly separates stochastic background heterogeneity from deterministic structural defects. This representation aligns with structural health monitoring needs, which focus on localizing anomalies against known geological backgrounds. The FNO surrogate acts as a high-fidelity alternative to traditional finite element methods. It accurately captures complex non-linear mappings with a mean relative error of 3.18% and achieves a 345 $\times$ speedup in forward modeling. This effectively removes computational bottlenecks in stochastic inversion. Guided by this surrogate and topological initialization, TG-FNO overcomes the issue of local minima stagnation commonly seen in baseline approaches. Repeated trials confirm its statistical stability and accuracy. The framework achieves a mean Inversion Success Rate (Acc_0) of 99.7% $\pm$ 0.6% and a high-fidelity reconstruction rate ($Acc_{0.5}$) of 89.3% $\pm$ 2.1%, outperforming conventional methods. Additionally, it maintains an overall 72 $\times$ computational acceleration over FEM-based inversion. This computational efficiency facilitates rapid routine monitoring, a critical requirement for timely decision-making in dam safety. The framework also shows robust out-of-distribution generalization. Without retraining, it accurately resolves incipient leakage signatures at a 10-fold contrast ratio and severe adverse conditions, including 0.20 Gaussian noise and 50% sensor data loss. This confirms that the learned operator approximates the underlying physical laws rather than merely memorizing statistical patterns. Current limitations define three primary research directions. First, explicit parameterization restricts detection to predefined shapes. Future work will adopt implicit representations, such as level set methods, for multi-anomaly scenarios. Second, the underlying Fourier transform is susceptible to spectral truncation. Integrating the synchrosqueezed wavelet transform (Li et al., 2017; Perez-Ramirez et al., 2016) will address this by improving localized feature resolution. Finally, the simulation-to-reality gap limits direct field deployment. Because neural operators approximate Green's functions, they remain highly sensitive to physical boundary variations. Future implementations must mitigate this by developing libraries of condition-specific operators to accommodate seasonal boundary shifts. Together, these advancements will build a practical foundation for operational digital twins in structural health monitoring.

Data availability

The dataset and source code supporting the findings of this study are publicly available in the TG-FNO Zenodo repository at https://doi.org/10.5281/zenodo.20305829.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The study was supported by the National Natural Science Foundation of China (Grant Nos. 52379097, 52509138), the Water Conservancy Science and Technology Project of Jiangxi Province (Grant No. 202426ZDKT27), the Guangzhou Association for Science and Technology (Grant No. QT-2025-049), and the Research and Innovation Program for Graduate Students in Chongqing (Grant No. CYB25282).

Appendix

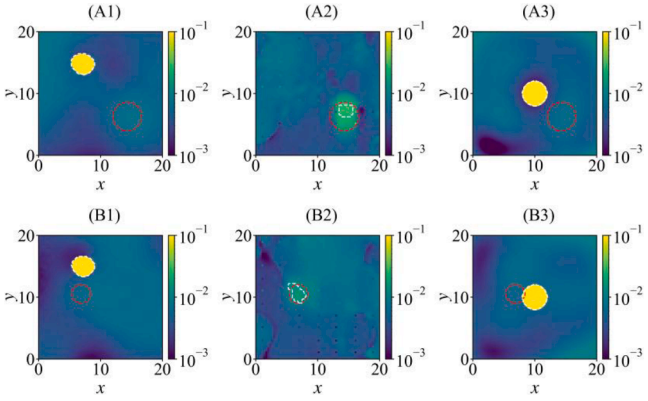


Fig. A1. Reconstruction performance of alternative baseline algorithms on representative realizations (R3 and R4). (A1–A3) Inverted conductivity fields for R3 using a Conditional GAN, Total Variation (TV) regularized pixel-wise inversion, and the Gauss-Newton (GN) algorithm, respectively; (B1–B3) Corresponding inverted fields for R4. Red dashed contours denote the true anomaly, while white dashed contours represent the reconstructed anomaly.

Table A1
Quantitative evaluation metrics of alternative baselines (CGAN, TV, and GN) justifying the baseline selection.

Evaluation Condition	Mean IoU	Acc_0	$Acc_{0.5}$	SSIM
Conditional GAN	0.0608	24%	0%	0.8952
Total Variation	0.1379	67%	7%	0.8675
Gauss-Newton	0.0629	40%	1%	0.8724

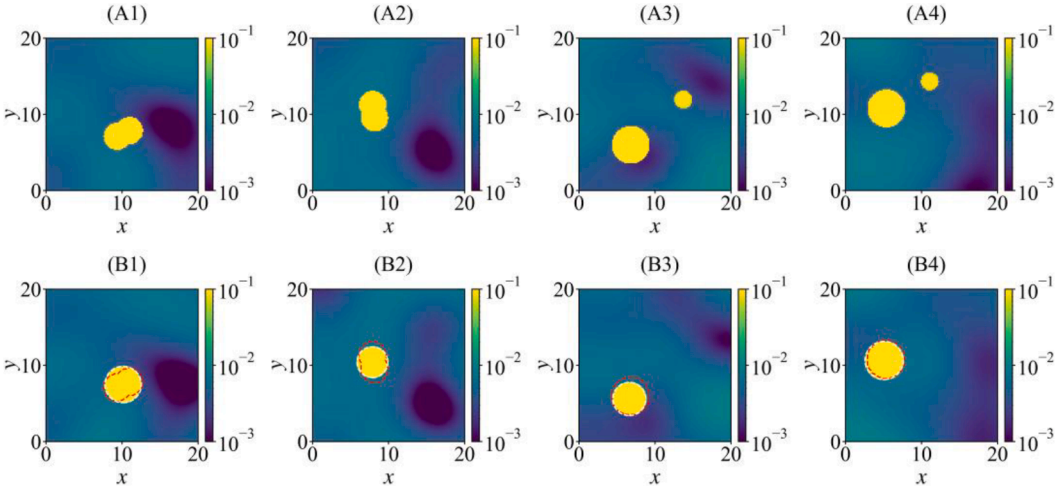


Fig. A2. Generalization of the proposed framework to out-of-distribution complex anomaly configurations. (A1–A4) True conductivity distributions featuring noncircular elongated profiles (A1, A2) and multiple leakage anomalies (A3, A4); (B1–B4) Corresponding inverted fields using the TG-FNO solver. Red dashed contours delineate the primary true anomaly, while white dashed contours denote the equivalent enclosing anomaly parameterized by the model.

References

Alkayem, N. F., Cao, M., Zhang, Y., Bayat, M., & Su, Z. (2018). Structural damage detection using finite element model updating with evolutionary algorithms: A survey. *Neural Computing and Applications, 30*(2), 389–411.

Ambikasaran, S., Li, J. Y., Kitanidis, P. K., & Darve, E. (2013). Large-scale stochastic linear inversion using hierarchical matrices: Illustrated with an application to crosswell tomography in seismic imaging. *Computational Geosciences, 17*(6), 913–927.

Amezquita-Sanchez, J. P., Valtierra-Rodriguez, M., & Adeli, H. (2018). Wireless smart sensors for monitoring the health condition of civil infrastructure. *Scientia Iranica, 25*(6), 2913–2925.

Bekele, B., Song, C., Abualshar, B., & Hunde, A. (2025). Real-time seepage health monitoring using spatial autocorrelation of distributed temperature data from fiber optic sensor. *Journal of Civil Structural Health Monitoring, 15*(4), 1101–1116.

Bekele, B., Song, C., Kim, S., & Eun, J. (2024). Seepage anomaly detection via Fiber optic distributed temperature sensing: Insights from physical and numerical modeling. *Journal of Geotechnical and Geoenvironmental Engineering, 150*(6), Article 04024039.

Bherde, V., Balunaini, U., Kambhammettu, B. V. N. P., & Regonda, S. (2025). Infrared thermography based seepage and erosion detection in earthen dams using laboratory scale models. *Scientific Reports, 15*(1), Article 23226.

Bonelli, S., & Brivois, O. (2008). The scaling law in the hole erosion test with a constant pressure drop. *International Journal for Numerical and Analytical Methods in Geomechanics, 32*(13), 1573–1595.

Cha, Y.-J., Ali, R., Lewis, J., & Büyüköztürk, O. (2024). Deep learning-based structural health monitoring. *Automation in Construction, 161*, Article 105328.

Chan, T. F., & Tai, X.-C. (2004). Level set and total variation regularization for elliptic inverse problems with discontinuous coefficients. *Journal of Computational Physics, 193*(1), 40–66.

Cimrman, R., Lukeš, V., & Rohan, E. (2019). Multiscale finite element calculations in Python using SfePy. *Advances in Computational Mathematics, 45*(4), 1897–1921.

Constable, S. C., Parker, R. L., & Constable, C. G. (1987). Occam's inversion: A practical algorithm for generating smooth models from electromagnetic sounding data. *Geophysics, 52*(3), 289–300.

Dong, H., Li, X., Yeh, T.-C. J., & Ge, J. (2020). Diagnosis of concentrated leakage channel embedded in dam base by means of hydraulic tomography. *Bulletin of Engineering Geology and the Environment, 79*(6), 2871–2881.

Entezami, A., Sarmadi, H., Kiarad, H., & Behkamal, B. (2026). Residual-informed deep meta regressor for predicting modal frequencies of historical masonry buildings under environmental and operational data scarcity. *Mechanical Systems and Signal Processing, 242*, Article 113675.

Ghanem, R., & Spanos, P. D. (1991). *Stochastic finite elements: A spectral approach*. New York: Springer-Verlag. https://doi.org/10.1007/978-1-4612-3094-6

Guo, Q., Zhao, Y., Lu, C., & Luo, J. (2023). High-dimensional inverse modeling of hydraulic tomography by physics informed neural network (HT-PINN). *Journal of Hydrology, 616*, Article 128828.

Guo, Y., Cui, Y., Xie, J., Luo, Y., Zhang, P., Liu, H., & Liu, J. (2022). Seepage detection in earth-filled dam from self-potential and electrical resistivity tomography. *Engineering Geology, 306*, Article 106750.

Haghighat, E., Raissi, M., Moure, A., Gomez, H., & Juanes, R. (2021). A physics-informed deep learning framework for inversion and surrogate modeling in solid mechanics. *Computer Methods in Applied Mechanics and Engineering, 379*, Article 113741.

Helton, J. C., & Davis, F. J. (2003). Latin hypercube sampling and the propagation of uncertainty in analyses of complex systems. *Reliability Engineering & System Safety, 81*(1), 23–69.

ICOLD. (2019). ICOLD incident database: Bulletin 99 update. *Statistical Analysis of Dam Failures*. Committee on Dam Safety, International Commission on Large Dams. Final Draft.

Ishizu, K., Goto, T.-N., Fukahata, Y., Koike, K., Vachiratienchai, C., & Siripunvaraporn, W. (2025). Inversion algorithm determining sharp boundaries in electrical resistivity tomography. *Geophysics, 90*(3), WA221–WA233.

Javadinasab Hormozabad, S., Gutierrez Soto, M., & Adeli, H. (2021). Integrating structural control, health monitoring, and energy harvesting for smart cities. *Expert Systems, 38*(8), Article e12845.

Ji, Y., Zha, Y., Yeh, T.-C. J., Shi, L., & Wang, Y. (2024). Groundwater inverse modeling: Physics-informed neural network with disentangled constraints and errors. *Journal of Hydrology, 640*, Article 131703.

Kaewnuratchadasorn, C., Wang, J., & Kim, C.-W. (2024). Neural operator for structural simulation and bridge health monitoring. *Computer-Aided Civil and Infrastructure Engineering, 39*(6), 872–890.

Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics, 3*(6), 422–440.

Kovachki, N., Li, Z., Liu, B., Azizzadenesheli, K., Bhattacharya, K., Stuart, A. M., & Anandkumar, A. (2023). Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research, 24*(89), 1–97.

Krishnapriyan, A., Gholami, A., Zhe, S., Kirby, R., & Mahoney, M. W. (2021). Characterizing possible failure modes in physics-informed neural networks. *Advances in Neural Information Processing Systems, 34*, 26548–26560.

Lee, K., & Carlberg, K. T. (2020). Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics, 404*, Article 108973.

Li, K., & Ye, W. (2025). D-FNO: A decomposed fourier neural operator for large-scale parametric partial differential equations. *Computer Methods in Applied Mechanics and Engineering, 436*, Article 117732.

Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A.M., Anandkumar, A., (2021). Fourier Neural Operator for Parametric Partial Differential Equations. International Conference on Learning Representations (ICLR), 2010.08895.

Li, Z., Park, H. S., & Adeli, H. (2017). New method for modal identification of super high-rise building structures using discretized synchrosqueezed wavelet and hilbert transforms: Super high-rise building structures. *Structural Design of Tall and Special Buildings, 26*(3), Article e1312.

Liu, B., Wang, C., Liu, Z., Xu, Z., Nie, L., Pang, Y., Wang, N., & Feng, S. (2021). Cascade surface and borehole geophysical investigation for water leakage: A case study of the Dehou reservoir, China. *Engineering Geology, 294*, Article 106364.

Loke, M. H., & Barker, R. D. (1996). Rapid least-squares inversion of apparent resistivity pseudosections by a quasi-Newton method. *Geophysical Prospecting, 44*(1), 131–152.

Long, D., Xu, Z., Yuan, Q., Yang, Y., & Zhe, S. (2025). Invertible fourier neural operators for tackling both forward and inverse problems. In *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics* (pp. 3043–3051).

Marty, T., Semet, Y., Auger, A., Héron, S., & Hansen, N. (2023). Benchmarking CMA-ES with basic integer handling on a mixed-integer test problem suite. In *Proceedings of the Companion Conference on Genetic and Evolutionary Computation* (pp. 1628–1635).

Molinaro, R., Yang, Y., Engquist, B., & Mishra, S. (2023). Neural inverse operators for solving PDE inverse problems. *Proceedings of the 40th International Conference on Machine Learning, Proceedings of Machine Learning Research, 202*, 25105–25139.

Moreno-Rodenas, A., Mantilla-Jones, J. D., & Valero, D. (2025). Age, climate and economic disparities drive the current state of global dam safety. *Nature Water, 3*(3), 284–295.

Nasser, R., Tzioutzios, D., & Liu, Y. (2025). Statistical analysis of global dam accidents in the 21st century: A focus on common features and causes. *Water Resources Management, 39*(12), 6493–6516.

Perez-Ramirez, C. A., Amezquita-Sanchez, J. P., Adeli, H., Valtierra-Rodriguez, M., Camarena-Martinez, D., & Romero-Troncoso, R. J. (2016). New methodology for modal parameters identification of smart civil structures using ambient vibrations and synchrosqueezed wavelet transform. *Engineering Applications of Artificial Intelligence, 48*, 1–12. https://doi.org/10.1016/j.engappai.2015.10.005

Quqa, S., Landi, L., & Loh, K. J. (2023). Crack identification using electrical impedance tomography and transfer learning. *Computer-Aided Civil and Infrastructure Engineering, 38*(17), 2426–2442.

Rafiei, M. H., & Adeli, H. (2018). A novel unsupervised deep learning model for global and local health condition assessment of structures. *Engineering Structures, 156*, 598–607.

Rahimi, M., Wood, C. M., Befus, K. M., & Rahimi, S. (2025). Challenges of 2D electrical resistivity tomography for detecting thin clay layers under levees: A case study of melvin price reach of the wood river levee on the mississippi river. *Near Surface Geophysics, 23*(3), 173–191.

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics, 378*, 686–707.

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. Cambridge, MA: MIT Press.

Saibaba, A. K., & Kitanidis, P. K. (2012). Efficient methods for large-scale linear inversion using a geostatistical approach. *Water Resources Research, 48*(5), Article W05522. https://doi.org/10.1029/2011WR011778

Taghizadeh, M., Nabian, M. A., & Alemazkoor, N. (2025). Multifidelity graph neural networks for efficient and accurate mesh-based partial differential equations surrogate modeling. *Computer-Aided Civil and Infrastructure Engineering, 40*(7), 841–858.

Tang, M., Liu, Y., & Durlofsky, L. J. (2020). A deep-learning-based surrogate model for data assimilation in dynamic subsurface flow problems. *Journal of Computational Physics, 413*, Article 109456.

Tiprak, K., Takeya, K., & Sasaki, E. (2025). Multi-restart CMA-ES with NNs and SNT for finite element model updating: A case of a short-span prestressed concrete girder bridge. In S. Tangtermsirikul, P. Warnitchai, K. Panuwatwanich, & G. Tanapornraweekit (Eds.), *725. Proceedings of the 18th East Asia-Pacific Conference on Structural Engineering and Construction, Volume 2* (pp. 468–475).

Wan, C. F., & Fell, R. (2004). Investigation of rate of erosion of soils in embankment dams. *Journal of Geotechnical and Geoenvironmental Engineering, 130*(4), 373–380.

Wang, S., Teng, Y., & Perdikaris, P. (2021). Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing, 43*(5), A3055–A3081.

Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing, 13*(4), 600–612.

Xiao, S., Yang, J., Ma, C., Li, P., Zhang, Z., Cheng, L., & Tong, F. (2024). Nondestructive testing of seepage in check dams using high-density electrical resistivity tomography based on laboratory test. *Construction and Building Materials, 411*, Article 134265.

Xue, J., Zhang, P., Li, J., Lu, X., Li, Z., Li, Y., Chen, J., & Kuang, C. (2025). A deep learning methodology for rapid identification of slab damage in concrete face rockfill dams. *Computer-Aided Civil and Infrastructure Engineering, 40*, 5596–5624.

Yin, Y., Deng, X., Hao, H., Hao, Y., Chang, H., & Yeh, T.-C. J. (2026). A graph fourier kolmogorov-arnold network (G-FourierKAN) and its application to spring discharge simulation. *Journal of Hydrology, 664*, Article 134313.

Zhang, B., Liang, Y., Wang, P., Dai, L., Jim Yeh, T., Zhang, H., Xu, B., Yin, C., Xia, R., & Sun, Z. (2026a). Characterizing concentrated leakage channels in full-scale earth

dam using geostatistical electrical resistivity tomography: An on-site case study. *Journal of Hydrology, 666*, Article 134756.

Zhang, D., & Lu, Z. (2004). An efficient, high-order perturbation approach for flow in random porous media via Karhunen–Loève and polynomial expansions. *Journal of Computational Physics, 194*(2), 773–794.

Zhang, W., Tian, C., Yang, J., Xiao, P., Wu, Z., Li, H., & Sun, W. (2026b). Vision and deep learning-based tunnel surface defect detection: Technological evolution, challenges, and prospects. *Measurement, 257*, Article 118853.